Problem 8.5

Calculate the present value of \$2000 payable in 10 years using an annual effective discount rate of 8%.

Solution

Under an annual effective discount rate $d = 0.08$, the present value of an amount due in n years is

$$\mathrm{PV} = F(1-d)^n.$$

Here, $F = 2000$ and $n = 10$, so

$$\mathrm{PV} = 2000(1-0.08)^{10} = 2000(0.92)^{10}.$$

Computing this value,

$$\mathrm{PV} \approx 868.7769.$$

Therefore, the present value is

$$\boxed{\$868.78}.$$